MASTER IAAA

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A High-Resolution Catch Reconstruction Model for Giant Squid (*Dosidicus gigas*)

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Contents

1 Introduction

The South-East Pacific marine ecosystem is distinguished by its remarkable primary productivity, within which the giant squid (*Dosidicus gigas*) occupies a pivotal role (Alegre et al. 2014). As an endemic pelagic predator, this cephalopod exerts strong top-down regulatory pressure on small pelagic fish whilst constituting an essential prey item for large marine vertebrates (Gong et al. 2020). Beyond its ecological function, *Dosidicus gigas* supports one of the world's major invertebrate fisheries, driving critical economic activity for international fleets (Arkhipkin et al. 2015).

Characterised by a short lifespan (rarely exceeding two years) and a high metabolic rate, the giant squid is highly susceptible to oceanographic variability (Nigmatullin et al. 2001). Its spatial distribution responds rapidly to global climate forcings, such as the El Niño-Southern Oscillation (ENSO), and to mesoscale hydrodynamic structures (fronts, upwellings, eddies) that facilitate prey aggregation (Waluda and P. G. Rodhouse 2006; Wu et al. 2025). Anticipating these spatial dynamics is therefore essential for the sustainable management of fish stocks.

Historically, the modelling of this spatial distribution has relied upon generalised additive models (GAMs) or neural networks applied to temporally smoothed data (Xie, Liu, Chen, Yu and Wang 2024). Although effective for monthly assessments, these models struggle to meet the current demands of operational oceanography, which necessitates forecasts on a weekly or daily scale (Yang et al. 2025).

Increasing the temporal resolution, however, highlights a structural limitation within conventional deep learning architectures (Xie, Liu, Chen, Yu, Wang and Xu 2025). Fisheries data are inherently noisy, asynchronous, and characterised by an exclusive presence-only bias (Gemmell et al. 2024). Training a daily supervised model frequently fails to correlate continuous environmental variables with discrete catch reports, which are mostly spatially aggregated. Consequently, standard approaches impose a 15- to 30-day smoothing period, thereby masking high-frequency ecological processes.

This dissertation proposes to address this issue through weakly supervised learning, drawing on the Multiple Instance Learning (MIL) paradigm (Ilse et al. 2018). Here, catches are treated as sets of high-frequency environmental observations, with the model tasked to isolate the most predictive physical anomalies within these sets. To stabilise the extraction of temporal features, the architecture incorporates self-supervised pre-training focused on fluid kinematics. This is complemented by regularisation via physics-informed neural networks (PINNs), ensuring that statistical predictions comply with the laws of hydrodynamics.

2 Material and Methods

2.1 Datasets

Habitat modelling requires the coupling of physical, biogeochemical, and anthropogenic variables. Four reference datasets were selected to inform these various dimensions.

2.1.1 Global Ocean Physics Reanalysis: GLORYS12V1

Hydrodynamic characterisation is provided by the GLOBAL_MULTIYEAR_PHY_001_030 (GLORYS12V1) product from the Copernicus Marine Environment Monitoring Service (CMEMS). This reanalysis provides three-dimensional ocean states at a 1/12° resolution (approximately 8 km) across 50 vertical levels (Drévillon et al. 2025).

The system is based on the NEMO model and incorporates a Kalman filter (SAM2) for the assimilation of in situ and satellite data (Lellouche, Greiner, Le Galloudec et al. 2018; Lellouche, Greiner, Bourdallé-Badie et al. 2021). The parameters extracted for this study include potential temperature (θ_o), salinity (S_o), zonal and meridional current velocities (u_o, v_o), sea surface elevation (zos), and mixed layer depth ($mlotst$).

2.1.2 Biogeochemical Dynamics: GlobColour

Primary production is estimated via chlorophyll-a concentration derived from the OCEANCOLOUR_GLO_BGC_L4_MY_009_104 product (Colella et al. 2026). To mitigate observational gaps caused by cloud cover, this product merges data from multiple multispectral sensors and applies spatio-temporal interpolation methods (objective kriging and DINEOF) (Saulquin et al. 2019; Alvera-Azc'arate et al. 2005). This provides an uninterrupted daily grid at a 4 km resolution.

2.1.3 Bathymetry: GEBCO 2025 Model

Seafloor morphology, which is decisive in the dynamics of water masses, is detailed by the GEBCO_2025 model. Published by the Seabed 2030 project, it offers a 15 arc-second resolution.

This model combines spatial altimetry data (SRTM15+) with high-precision acoustic bathymetric surveys. The dataset includes a Type Identifier (TID) grid, which specifies the source of information for each pixel. This metadata allows for the distinction between measured and extrapolated depths, providing a means to spatialise uncertainty in bottom friction modelling.

2.1.4 Fishing Effort: Global Fishing Watch

Extractive pressure is quantified through the apparent fishing effort grids from the Global Fishing Watch initiative, at a high spatial resolution of $0.01°$ (Watch 2025).

Effort is calculated by analysing the AIS signals emitted by industrial fishing vessels. Processing via convolutional neural networks (CNNs) is performed in two stages: the first network infers the gear type, whilst the second classifies vessel behaviour, distinguishing transit phases from active extraction operations (Kroodsma et al. 2018). The spatialised effort density will be used to represent the global footprint of local fishing activities within the environmental strata.

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